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Project Title:	miRNA profiling of radiation response: A systems biology approach to understanding regulation of proton and heavy ion dose effects		
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Flight Program:			
Flight Assignment:	NOTE: received NCE until 9/29/2011 per PI (08/2010)		
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Task Description:	During space missions, astronauts will be exposed to radiation. Characterization of the space radiation environment is well established procedure which involves various physical dosimeters capable of measuring large spectrum of radiation energies. However, these dosimeters don't show the effect of radiation on astronaut's health. Measuring of these effects is achieved using biomarkers of ionizing radiation exposure. In contrast with physical dosimeters, radiation biomarkers have the potential to provide information on both radiation dose and radiation-induced damage or changes of biological functions. In space flights, biomarkers could be used to evaluate radiation induced cell damage and the physiological status of astronauts including recovery or disease development. In our work, we took advantage of two powerful research tools. The first one was high-throughput screening, a modality that allows researchers to quickly conduct millions of genetic tests. Our work involves such an approach to monitor radiation induced changes of gene expression in cells from both mouse models and human subjects. The second important factor was the accessibility of "space" type of high energy radiation which is generated in the NASA Space Radiation Laboratory (NSRL) located at Brookhaven National Laboratory. We used NSRL for most of our radiation experiments. Specifically, we monitored the expression of two classes of genes: miRNA coding genes and protein coding genes. Since miRNA are very important meta-regulators of gene expression, we suspected that monitoring of the changes in their expression levels could reveal very important information about the radiation conditions and the health of cells and organisms. The second class of genes examined were the protein coding genes for which we had preliminary information implicating them in radiation response. We measured gene expression changes in blood of mice and in humans. The radiation that we used in the experiments with mice was a simulation of space radiation that would be encountered by astronauts in low earth orbit and in the interplanetary space. We also measured miRNA and gene expression in human blood samples that were provided to us by consenting radiotherapy patients. Here we present the overall achievements of our work.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research may have significant impact on our understanding of human response to radiation. As we show, radiation induces robust miRNA and gene expression changes in blood cells. Importantly, the characterization of this response can be used as indicator of the radiation induced functional changes in blood cells, including the length of the response and recovery. In addition, radiation induced miRNA signatures can be used for development of radiation classifiers which can show retrospectively the radiation dose of astronauts, radiation workers or victims of nuclear and radiation accidents.
	SUPPLEMENTAL REPORTING FOR FINAL REPORT--DECEMBER 2011 Goals: The main goals of our studies are to generate radiation specific miRNA and gene expression signatures from cultured cells and whole blood samples of human and mouse origin. The scope of the study includes evaluation of these signatures for use in two modalities: first, as a tool for better understanding of the radiation induced functional changes in cells and organisms and second, as a biomarker for space and occupational radiation exposure. These results will lead to better understanding of the radiation induced functional changes in living organisms. Summary of the results: Mouse studies: This study investigates to what extent miRNA expression signatures in peripheral blood can be used as biomarkers for low- and high-LET radiation exposures. Mice were irradiated with various doses of gamma-rays and high LET 56Fe-ions and protons. Blood was collected at 6 h and 24 h after irradiation. Differentially expressed and dose-dependent miRNAs were determined by means of real-time PCR and used to build various class predictors and to perform gene-ontology and network analyses. Our main finding is that miRNA expression signatures derived from mouse blood are specific for every irradiation condition we studied. We used the detected differentially expressed miRNA, to identify biological processes and functions potentially impacted by radiation. This was done by miRNA target analysis in which we determined the protein coding genes that are targeted by the differentially expressed miRNA. These genes were subject to gene ontology analysis in order to identify the functional changes induced by radiation. Overall the analysis show that radiation induces changes in miRNA control of specific biological processes and molecular functions, such as DNA binding, transcription factor activity, metabolic processes, and others. There is a large overlap between the genes targeted in the gamma-, proton-, and 56Fe ion-irradiated blood samples. Accordingly, a similar pattern of signal transduction developed as evidenced by the similarities in gene ontology processes and functions. While the condition specific sets of genes are not identical, there is sufficient commonality, which indicates a unified overall response to radiation. The evident differences among the target genes and GO categories, depending on irradiation type, point to a precise discrimination of the damage, induced by the different types of radiation. A practical outcome of our study is that it is possible to develop classifiers of radiation exposure. Based on the detected differences in miRNA expression signatures, we developed statistical classifiers that can be used to identify the type of radiation and the dose that mice were exposed. Our predictors classified the samples quite well according to radiation type (accuracy of 81%) and dose (accuracy of between 75-100% depending of the type of radiation). These classifiers encompass various scenarios that require predictions about received radiation exposures. It is possible to run two predictors in sequence. For example, if neither the radiation type nor the dose of exposure is known, the classifier predicting radiation type can be applied first, and, after the radiation type has been determined, a gamma-rays, proton-, or 56Fe-ion specific classifier can be used to estimate the dose of exposure. An important advantage is that the classifiers do not require a pre-exposure control sample because 1) all samples that were used to develop the classifiers were derived from separate, independent organisms and 2) the nearest-centroid classifiers that we developed are able to assign a radiation type and dose to a sample with unknown irradiation status based on the absolute normalized expression values of the miRNA contained in the classifier. Human studies: The goal of these studies was to determine the effect of radiation on miRNA and gene expression in human blood in vivo. We obtained blood under IRB-approved protocols from 16 patients that were subject to total body irradiation with 1.25 Gy for therapeutic purposes. Samples were collected prior to irradiation and 4 hours afterward. All patients were in complete remission stage 1 or 2 with no blast cells detected in the peripheral blood. None of the patients had prior radiation treatment or had been treated with genotoxic drugs for at least 5 weeks before the start of radiation treatment. We tested to what degree gene expression in the blood cells of these patients comparable with the one is obtained from healthy individuals, by comparison of the gene expression patterns of 4 tumor patients and 4 normal healthy individuals. The results showed strong overlap between these two groups. This means that the expression profiles obtained in this study represent healthy cells of cancer patients in complete remission, and not responses in cancer cells.

Task Progress:

The comparison of the miRNA and gene expression between eight pre- and post-irradiated subjects revealed significant changes Overall, 45 miRNAs (23% of all expressed miRNA) show statistically significant expression changes after irradiation. Very importantly, despite the genetic differences, 27 of these miRNAs were upregulated in all patients. Additionally, gene expression analysis show statistically significant changes in the expression levels of 275 protein coding genes. The main conclusion is that radiation induces changes in the expression levels of a large set of miRNAs, a substantial portion of which are common among all participating individuals. This supports the idea of a robust miRNA response induced by radiation in blood cells. While gene expression changes in patients after irradiation were studied before, this study is the first one to determine miRNA expression changes in human blood cells in vivo.

To determine the effect of the overexpressed miRNAs on cell functions we performed a Gene Ontology analysis of the miRNA target genes. Our results indicate that miRNAs exert control on about 30% of the biological processes induced by radiation. Specifically, irradiation of blood cells results in an increased miRNA control of genes involved in hemopoiesis, immune response, signal transduction, response to stress, cell cycle and apoptosis. These results indicate that miRNA and gene expression profiles could be used as post irradiation functional biomarkers in humans.

We build also class-prediction classifiers to test to what degree class prediction is able to indicate the irradiation status of human blood samples based on their miRNA expression profiles. The classifiers containing all 45 differentially expressed miRNAs performed with an accuracy of 100% in all classification methods used, showing the high potential of miRNA signatures in radiation biodosimetry.

In summary, our studies show that radiation induces robust miRNA response in blood cells. This response is specific to the type of radiation and the radiation dose. The specificity of miRNA expression signatures can be used for dose estimation and for monitoring of the changes in health of radiotherapy patients, radiation workers and astronauts in space.

ANNUAL PROGRESS REPORT--AUGUST 2010

Goals: The main goal of this study is to generate radiation specific miRNA expression signatures from cultured cells and whole blood samples of human and mouse origin. The scope of the study includes evaluation of these signatures for use in two modalities: first, as a biomarker for space and occupational radiation exposure and second, as a tool for better understanding of the functional changes in biology induced by radiation.

Summary of the results:

Mouse studies:

This study investigates to what extent microRNA (miRNA) expression signatures in peripheral blood can be used as biomarkers for low- and high-LET radiation exposures. Mice were irradiated with various doses of gamma-rays or 56Fe ions and blood was collected at 6 h and 24 h after irradiation. Differentially expressed and dose-dependent miRNAs were determined by means of real-time PCR and used to build various class predictors and to perform gene-ontology and network analyses.

Our main finding is that miRNA expression signatures derived from mouse whole blood are specific for every irradiation condition we studied. Of 320 miRNAs examined, a total of 35 were differentially expressed (estimated False Discovery Rate (FDR) <7%, $p<0.001$). The expression data were used to generate predictive classifiers. When applied to miRNA samples with unknown irradiation status, these classifiers allow for identification of the type and dose of radiation to which the samples were exposed. We developed three types of classifiers: 1) a class predictor designed to distinguish between radiation type (gamma, 56Fe 1GeV, or control). This predictor correctly classified 75% of the samples. 2) a class predictor for dose irrespective of time point after radiation exposure. The gamma predictor correctly classified 83% of the samples; the Fe predictor correctly classified 100% of the samples. 3) Four radiation type- and time point-specific classifiers were built. The gamma, 6 h and 24 h predictors correctly classified 92% and 89% of the samples, respectively. The 56Fe-ions irradiation predictors (Fe, 7 h and Fe, 24 h) correctly classified 100% of the samples. An advantage of our classifiers is that they encompass various scenarios that require predictions about received radiation exposures. It is possible to run two predictors in sequence. For example, if neither the radiation type nor the dose of exposure is known, the classifier predicting radiation type can be applied first, and, after the radiation type has been determined, a gamma- or 56Fe-specific classifier can be used to estimate the dose of exposure. A further advantage of our classifiers is that they do not require a pre-exposure control sample because all samples that were used to develop the classifiers were derived from separate, independent organisms. The analysis of the miRNA expression levels in the proton irradiated mice are in progress.

Our next goal was to show to what extent miRNA expression signatures can be used as biomarkers for the biological effects of radiation. Since each of the irradiation conditions we used was characterized by the expression of specific sets of miRNAs, it was of great interest to identify what processes were controlled by these sets. The results from the Gene Ontology analyses (done under statistically very restrictive conditions) show that after irradiation miRNAs increase their targeting (control) of genes related to specific biological processes. A comparison between the Gene Ontology categories controlled in different irradiation conditions reveals important similarities and differences. For example, miRNA control of one group of biological processes (we can name it “core processes”) was increased 6 h or 7 h after irradiation in all irradiation conditions. Remarkably these Gene Ontology categories are mRNA transcription; mRNA transcription regulation; nucleoside nucleotide and nucleic acid metabolism; and developmental processes. This points to an increased miRNA control of cell reprogramming since at least two of these categories contain many transcription factors. Other Gene Ontology categories were dose-dependent. Higher doses led to an increased miRNA control of genes involved in the cell cycle and in signal transductions and of other genes related to the cellular stress response. Notably, the number of regulated Gene Ontology categories increased with increased radiation-induced damage. All this shows a mode of control in which a group of core processes (common to all doses) and a group of dose-specific processes are targeted by miRNAs after irradiation. Together, our findings from the mouse studies illustrate that miRNA mediated control of gene expression could be very important for estimating radiation-induced damage and the degree of damage repair.

Human studies:

miRNA and gene expression changes in blood cells of radiotherapy patients. The goal of these studies was to determine the effect of radiation on miRNA and gene expression in human blood cells in vivo. We obtained blood under IRB-approved protocols from 16 patients that were subject to total body irradiation with 1.25 Gy for therapeutic purposes. Samples were collected prior to irradiation and 4 hours afterward. All patients were in complete remission

stage 1 or 2 with no blast cells detected in the peripheral blood. None of the patients had prior radiation treatment or had been treated with genotoxic drugs for at least 5 weeks before the start of radiation treatment. We tested to what degree gene expression in the blood cells of these patients is comparable with the one obtained from healthy individuals, by comparison of the gene expression patterns of 4 tumor patients and 4 normal healthy individuals. The results showed strong overlap between these two groups. This means that the expression profiles obtained in this study represent healthy cells of cancer patients in complete remission, and not responses in cancer cells.

miRNA and gene expression results

The comparison of the miRNA and gene expression between eight pre- and post-irradiated subjects revealed significant changes. Despite the genetic differences between the patients, 45 miRNAs (23% of all expressed miRNA) (FDR < 8%, average p-value of 0.006, fold change > 2) and 254 genes ($p < 1.10\text{-}5$ and FDR < 4%), show statistically significant expression changes after irradiation. Very importantly, 27 of these miRNAs were upregulated in all patients. The main conclusion from determining the miRNA expression changes in response to TBI is that radiation induces increases in the expression levels of a large set of miRNAs, a substantial portion of which are common among all participating individuals. This supports the idea of a robust miRNA response induced by radiation in blood cells. While gene expression changes in patients after irradiation were studied before, this study is the first one to determine miRNA expression changes in human blood cells in vivo. Class prediction

We build class-prediction classifiers to test to what degree class prediction is able to indicate the irradiation status of human blood samples based on their miRNA expression profiles. The leave-one-out cross validation method was used to test the robustness of the classifiers. In this method, all sample pairs—except one—are used to build the classifier, and the ability of the classifier to correctly predict the class membership of the left-out sample pair is computed. This process is continued until each pair has been left out once.

The classifiers containing all 45 differentially expressed miRNAs performed with an accuracy of 100% in all classification methods used, showing the high potential of miRNA signatures in radiation biodosimetry.

Functional analysis of miRNA and gene expression

The goal of the functional analysis is to determine the effect of the overexpressed miRNAs on cell functions. The downregulated genes targeted by the radiation-induced miRNAs, as well as all differentially expressed genes, were uploaded to the PANTHER v. 7.0 Gene Ontology database, and gene-set enrichment analysis was performed in the ‘biological process’ category. Our results indicate that miRNAs control about 30% of the biological processes induced by radiation. Specifically, irradiation of blood cells results in an increased miRNA control of genes involved in hemopoiesis and the immune response, whereas metabolic processes are underrepresented among the genes controlled by miRNAs. A Gene Ontology analysis of the genes induced by radiation shows that they are involved in pathways regulating the immune response, signal transduction, the response to stress, the cell cycle, apoptosis, and hemopoiesis.

All these results indicate that miRNA and gene expression profiles could be used as post irradiation functional markers in humans.

In summary, our study shows that radiation induces a robust miRNA expression response in blood cells. These signatures may be used in the development of reliable radiation classifiers and as indicators for the changes in the processes active in blood cells after radiation exposure. This can be used for dose estimation and for monitoring the health of radiotherapy patients, radiation workers and astronauts in space.

Bibliography Type:	Description: (Last Updated: 08/25/2020)
Articles in Peer-reviewed Journals	Templin T, Amundson SA, Brenner DJ, Smilenov LB. "Whole mouse blood microRNA as biomarkers for exposure to gamma-rays and (56)Fe ion." Int J Radiat Biol. 2011 Jul;87(7):653-62. http://dx.doi.org/10.3109/09553002.2010.549537 ; Epub 2011 Jan 28. PubMed PMID: 21271940 , Jul-2011
Articles in Peer-reviewed Journals	Templin T, Paul S, Amundson SA, Young EF, Barker CA, Wolden SL, Smilenov LB. "Radiation-induced micro-RNA expression changes in peripheral blood cells of radiotherapy patients." Int J Radiat Oncol Biol Phys. 2011 Jun 1;80(2):549-57. Epub 2011 Mar 21. PubMed PMID: 21420249 , Jun-2011
Articles in Peer-reviewed Journals	Young EF, Smilenov LB. "Impedance-based surveillance of transient permeability changes in coronary endothelial monolayers after exposure to ionizing radiation." Radiat Res. 2011 Oct;176(4):415-24. Epub 2011 Aug 23. PubMed PMID: 21861763 , Oct-2011
Articles in Peer-reviewed Journals	Templin T, Young EF, Smilenov LB. "Proton radiation-induced miRNA signatures in mouse blood: Characterization and comparison with (56)Fe-ion and gamma radiation." Int J Radiat Biol. 2012 May 22. [Epub ahead of print] PubMed PMID: 22551419 , May-2012